

EFFECTS OF THREE ORAL ANALGESICS ON POSTOPERATIVE PAIN FOLLOWING ROOT CANAL PREPARATION: A CONTROLLED CLINICAL TRIAL

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INTRODUCTION

Pain is the most common reason for a visit to the dentist¹, and one of the most important objectives of root canal treatment is to alleviate this pain. Root canal treatment in itself is not a painful procedure if carried out under proper local anesthesia, ensuing postoperative pain is a concern for both the patient and the clinician. Post endodontic pain is intense during the first 24-48 hours after the procedure, reducing in intensity with the passage of time². This pain can be related to the amount of extruded debris from the canal into the periradicular tissues during the procedure. It can also be due to over-instrumentation or restoration in hyperocclusion³. There is also a positive correlation between post-endodontic pain and presence of preoperative pain and apprehension⁴.

Commonly used methods to relieve post-operative endodontic pain are administration of long acting local anesthetic agents and use of analgesic drugs. Some of these drugs act by preventing the production of inflammatory mediators called prostaglandins, thus decreasing pulpal and corresponding peri-radicular changes. Increased prostaglandin levels are linked with increased levels of inflammation and pain⁵.

Others have central and peripheral nervous system effects. Whether NSAIDS, Acetaminophen or opioids, all can be used to reduce pulpal and peri-radicular pain levels. Ibuprofen is the most commonly used analgesic with the safest profile⁶. Naproxen has longer duration of action as compared to ibuprofen but takes more time to reach similar plasma levels⁷. Tramadol is a non-narcotic opioid that acts on central nervous system to produce analgesia⁸.

Knowing an effective pharmacologic regimen to control post-operative endodontic pain has been the objective of many previously conducted studies⁹⁻¹¹, yet no definitive analgesic and anti-inflammatory protocol have been established. The present study compared the effects of Ibuprofen, Naproxen and Tramadol in controlling post-operative pain following pulpectomy in teeth with signs and symptoms of irreversible pulpitis.

METHODOLOGY

This quasi-experimental study was carried out from January to October 2018 in the Department of Endodontics, Fatima Jinnah Dental College and Hospital, Karachi. After approval from the Institutional Ethics Review board via certificate no. MAR-2018-OPR01, data was collected from randomly selected 120 subjects coming to the Department of Endodontics with the complaint of irreversible pulpitis. Previously

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reported means of 5.8 ± 2.24 (baseline), 0.7 ± 0.99 (after 24 hours) for group 1 and 5.8 ± 2.69 (baseline), 0.4 ± 0.97 (after 24 hours) for group 2 were used to calculate sample size by WHO calculator at 95% confidence interval. After explanation and comprehensive medical history, written consent was taken before the start of the procedure. Subjects were divided randomly into four groups of 30 each by using software Microsoft Excel 2013. A control group received a multivitamin medication as placebo and three test groups received a single dose of either tablet (400 mg of Ibuprofen), tablet (100 mg of Tramadol HCL or tablet (550 mg of Naproxen Sodium; ICI, Pakistan) immediately after the pulpectomy appointment.

A questionnaire was designed to record the VAS score before and after 6, 12 and 24 hours of treatment. The questionnaire contained 4 lines of 10 cm each drawn at some distance to allow patients to draw lines parallel to them according to their pain levels at specified time intervals.

Inclusion criteria was cooperative patients of both genders and age group of 18-60 years with spontaneous pain of moderate to severe nature associated with irreversible pulpitis and no signs or symptoms of apical periodontitis. Patients having no other tooth with symptoms of irreversible pulpitis in the same quadrant was also a criterion for inclusion.

Exclusion criteria was patients who had taken analgesics in the previous 6-12 hours, patients with previous bad dental experience, history of allergy to the experimental drugs, pregnant or lactating women, periodontally compromised tooth and significant medical history.

On the day of the appointment, Local anesthesia was administered using 1.8 ml cartridge of Lidocaine 1; 100,000 (Medicaine) and 27-gauge needle. Isolation was obtained by using rubber dam and access was gained into the pulp chamber. Electronic apex locator (Dentaport ZX; J. Morita, Japan) was used to determine working length and patency was achieved using K-file # 10. Further instrumentation was carried out with K-files # 15, 20 and 25 to achieve pulpectomy. Irrigation was done between instrumentation using 2.5% sodium hypochlorite. After pulpectomy had been carried out, access was sealed with cotton pellet and temporarily restored with Cavit (3M ESPE, St Paul, USA).

A coded sealed pack containing a randomly allocated tablet was given to the patient after the appointment and was instructed to take it immediately. A questionnaire containing biodata, VAS and secret code

of the medicine was handed over to the patient and was instructed to record the pain levels at specified intervals. Filled questionnaires were collected from the patients at the next appointment.

LIMITATIONS

However, there were certain limitations to this study. After taking the first dose in front of the operator, it could not be ensured that the patients had taken all subsequent doses at their specified time. Secondly pain perception varies between individuals¹⁸. This can be attributed to emotional status, psychological factors and difference in pain threshold levels. Furthermore considering the side effect profile of the prescribed drugs, risk to benefit ratio of these drugs has not been assessed. Further studies need to be carried out to overcome these limitations.

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CONCLUSION

This study concludes that Naproxen, Ibuprofen and Tramadol were equally effective in reducing post-instrumentation pain in the first 24 hours after root canal treatment in teeth with irreversible pulpitis.

CONFLICT OF INTEREST

This study has no conflict of interest to be declared by any author.

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